

a) $\mu(\bigcap_{n \in \mathbb{N}} A_n) = \lim_{n \rightarrow \infty} (\mu(A_n))$?

$$\mu(\bigcap_{n \in \mathbb{N}} A_n) = \mu(\emptyset) = 0 \quad \lim_{n \rightarrow \infty} (\mu(A_n)) = \infty$$

so, MCFA does not hold for $\{A_n\} \subset \mathcal{F}$.

b) Need something so that $\mu(A_n) < \infty \exists A_n$

let $\mu(A_n) = \frac{1}{|A|}$, then $\mu(\bigcap_{n \in \mathbb{N}} A_n) = \mu(\emptyset) = 0$ and
 $\lim_{n \rightarrow \infty} (\mu(A_n)) = 0$, so MCFA holds.

c) We need some n^* s.t. $\mu(A_{n^*}) < \infty$.

d) MCFA does not have a mandatory condition. It holds for all sequences of increasing sets. ✓

$$2) \Omega_0 \subset \Omega \quad \Omega_0 \notin \mathcal{F}$$

$$\mathcal{F}_0 = \{F \cap \Omega_0 : F \in \mathcal{F}\}$$

Ω_0 largest set in $\mathcal{F}_0$!

① $\emptyset \in \mathcal{F}$, so for $F = \emptyset$, $F \cap \Omega_0 = \emptyset$, so $\emptyset \in \mathcal{F}_0$ and when we show closure under complementation, $\Omega_0^* \in \mathcal{F}_0$ will be true too.

② Closure under Complementation

Take $A \in \{F \cap \Omega_0 : F \in \mathcal{F}\}$. $A \neq \Omega_0$ b/c $\Omega_0 \notin \mathcal{F}$.

~~So, anything in $A^c \cap \Omega_0$, meaning there is some B for every A s.t.~~

$$B = \Omega_0 \setminus A.$$

Also, for any $F \in \mathcal{F}$, $F^c \in \mathcal{F}$ since $\mathcal{F}$ is a σ -algebra.

If $F \cap \Omega = \emptyset$, then $F = \Omega \cap \Omega_0 = \Omega_0$ since $\Omega_0 \subset \Omega$.

So, for $A = \emptyset$, $A^c \in \mathcal{F}_0$.

If $F \cap \Omega_0 \neq \emptyset$, Any F s.t. $F \cap \Omega_0 = \Omega_0$ is also closed under complementation b/c $\emptyset \in \mathcal{F}_0$.

For $F \cap \Omega_0 \neq \emptyset$ and $F \cap \Omega_0 \subset \Omega_0$, let

$F \in \mathcal{F}$, so $F^c \in \mathcal{F}$. If $F \subset \Omega_0$, then $F^c \not\subset \Omega_0$, but

$F^c \cap \Omega_0 \in \mathcal{F}_0$, and Ω_0 is largest set in $\mathcal{F}_0$.

If $F \not\subset \Omega_0$, then $F^c \subset \Omega_0$ and

$F \cap \Omega_0 \in \mathcal{F}_0$.

So, $\mathcal{F}_0$ is closed under complementation!

Sloppy!
Sorry about
wording and
path.

Continued on
next page!

2 cont

(2)

(iii) Countable unions

$\mathcal{L}$ is closed under countable unions, and

Everything in $\mathcal{F}_0 \in \mathcal{F}$, so $\mathcal{F}_0$ is closed under
countable unions, x

-4

Identity $f \equiv 1$

a) $I(A)$ where ~~A~~ A is a non-borel measurable set.

b) ~~Non measurable~~
~~be above~~

like in a), except make 0 into -1.

Then

$$f(x) = \begin{cases} 1 & x \in A \\ -1 & x \notin A \end{cases}$$

A is not borel measurable,

f is not borel measurable, but clearly,

$|f| = 1$, which is borel measurable

4) $\mathcal{F}_1 = \{\emptyset, (-\infty, 5], (5, \infty), \mathbb{R}\}$ $\Omega_1 = \mathbb{R}$
 $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$

a) $T: \Omega_1 \rightarrow \Omega_2$

$$T(x) = \begin{cases} 5 & x \leq 0 \\ 0 & x > 0 \end{cases}$$

$T(x)$ is not $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable.

~~$T^{-1}(A) = \begin{cases} \emptyset \\ (-\infty, 0) \end{cases}$~~

~~Take $A \in \mathcal{F}_2$~~ s.t. $0 \in A$. E.g. $(-1, 1)$.
and 5 not $\in A$

Then, $T^{-1}(A) = (0, \infty)$. $\notin \mathcal{F}_1$. $\times$

b) $\varepsilon_1, \varepsilon_2, \dots$ are random variables, on $(\Omega, \mathcal{F}, P)$
 $\{X_i\}_{i=1}^\infty$ is a sequence of real numbers w/ $\sum X_i^2 < \infty$.

$$Y_i(\omega) = X_i \beta + \varepsilon_i(\omega) \quad \text{for some } \beta \in \mathbb{R}.$$

X_i is a sequence of real numbers, and $\beta \in \mathbb{R}$. So $X_i \beta \rightarrow \beta X_1, \beta X_2, \beta X_3, \dots$
Each βX_i is measurable, so $(\beta X_1, \beta X_2, \dots)$ is measurable (By Theorem).

So $X_i \beta$ is measurable.

Now, $\varepsilon_i(\omega)$ are random variables, so they are measurable.

By theorem, sum of two measurable functions is measurable, so

$$Y_i(\omega) = X_i \beta + \varepsilon_i(\omega) \text{ is measurable.}$$

We showed $Y_i(\omega)$ is measurable, and we know $X_i \in \mathbb{R} \forall i$, so $X_i Y_i(\omega)$ is measurable (finite products part of Theorem). (4)

$\sum X_i Y_i(\omega)$ is just a sum of measurable parts, so $\sum X_i Y_i(\omega)$ is measurable.

Since X_i is measurable, $X_i \cdot X_i = X_i^2$ is measurable, and hence,

$$\text{so is } \sum_{i=1}^n X_i^2.$$

Now, we proved in a homework problem that $\frac{f}{g} I(g \neq 0)$ is measurable for measurable f and g .

The problem told us that $\sum_{i=1}^n X_i^2 > 0$, so $I(g \neq 0)$ is unnecessary here.

Then

$$\frac{\sum_{i=1}^n X_i Y_i(\omega)}{\sum_{i=1}^n X_i^2} I(\sum X_i^2 \neq 0) = \frac{\sum_{i=1}^n X_i Y_i(\omega)}{\sum_{i=1}^n X_i^2} \text{ is measurable.}$$

So, $Y_i(\omega) = X_i \beta + \varepsilon_i(\omega)$ and $\hat{\beta}_n(\omega) = \frac{\sum_{i=1}^n X_i Y_i(\omega)}{\sum_{i=1}^n X_i^2}$ are

sequences of real valued, measurable functions.